

TrES-2b

R-2456

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120718 · created 2026-08-23T00:27:53+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456126.8343 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1491 +/- 0.001
Transit depth (Rp/Rs)^2	2.22 +/- 0.031 [%]
Orbital Inclination (inc)	84.77 +/- 0.43
Airmass coefficient 1 (a1)	0.023 +/- 0.013
Airmass coefficient 2 (a2)	2.9998 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	35.23 %
Best Comparison Star	None
Optimal Aperture	3.93
Optimal Annulus	13.89
Transit Duration (day)	0.0819 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1491 ± 0.001 vs 0.1254 (deviation 1.07σ)
Duration fit vs published	0.0819 d vs 0.0746 d (deviation 0.13σ)
Mid-time O–C	0.4 min
Depth SNR	6.7
Residual scatter	35.23 %

Light curve

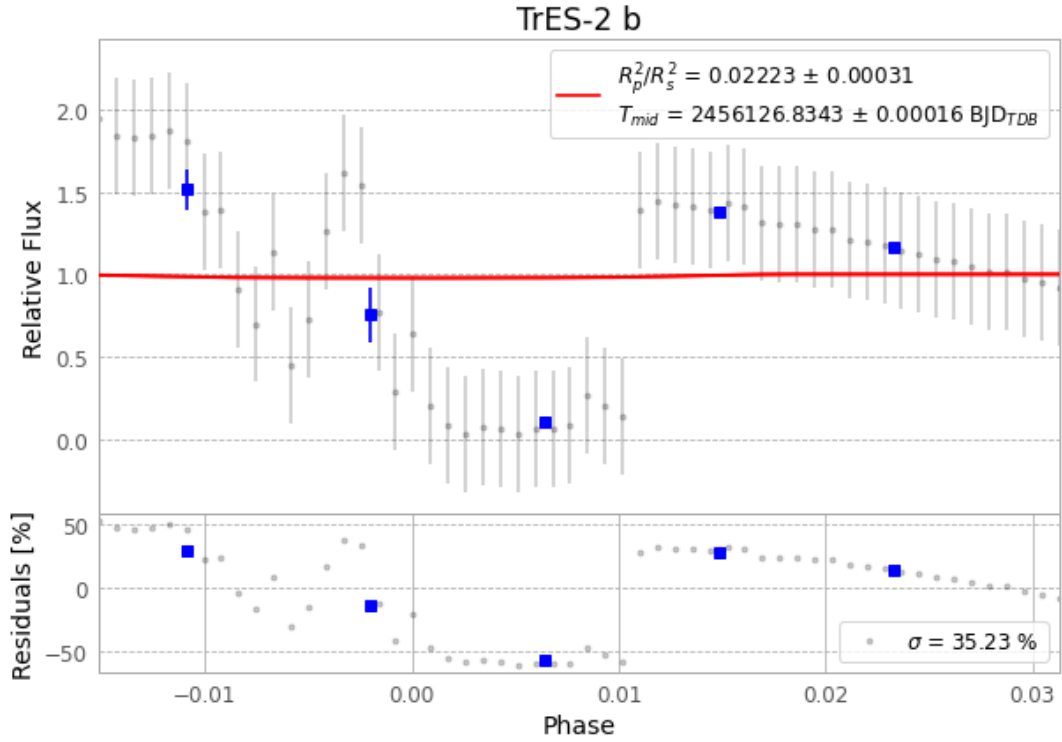


Figure 1 — FinalLightCurve_TrES-2 b_2012-07-18.png (SHA-256 0cc1b61ee654e06d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 18b90a181f1128d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), TRES-2120718070313.FITS → TRES-2120718094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120718.log (477e9505f6ad...), FinalLightCurve_TrES-2 b_2012-07-18.png (0cc1b61ee654...), FinalLightCurve_TrES-2 b_2012-07-18.csv (45a4f9a50c0d...)

Compute resources

Wall time 236.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:27Z.